

JAVA PROGRAM REPORT

Multiple Interfaces Implementation

Programming Language: Java

Topic: Implementation of Multiple Interfaces

Class Name: Multiple

Objective

To develop and execute a Java program that demonstrates multiple inheritance through interfaces. The class Multiple implements both I1 and I2 and provides implementations for all abstract methods.

Java Source Code

```
package core_java;

interface I1{
    int a = 10 ;
    void student1();
    void student3();
}

interface I2{
    double b = 30 ;
    void student2();
}

public class Multiple implements I1,I2{

    public void student2(){
        System.out.println(" i am student 2");
    }

    public void student1(){
        System.out.println("i am student 1");
    }

    public void student3(){
        System.out.println(" i am student 3");
    }

    public static void main(String[] args){
        Multiple m = new Multiple();
        m.student1();
        m.student2();
        m.student3();
        System.out.println(m.a);
    }
}
```

Program Explanation

Interface I1: Contains an integer variable a with value 10 and declares the methods student1() and student3().

Interface I2: Contains a double variable b with value 30 and declares the method student2().

Multiple Class: The class implements both I1 and I2. Therefore, it provides implementations for student1(), student2(), and student3().

Main Method: An object m of the Multiple class is created. The three methods are called using this object.

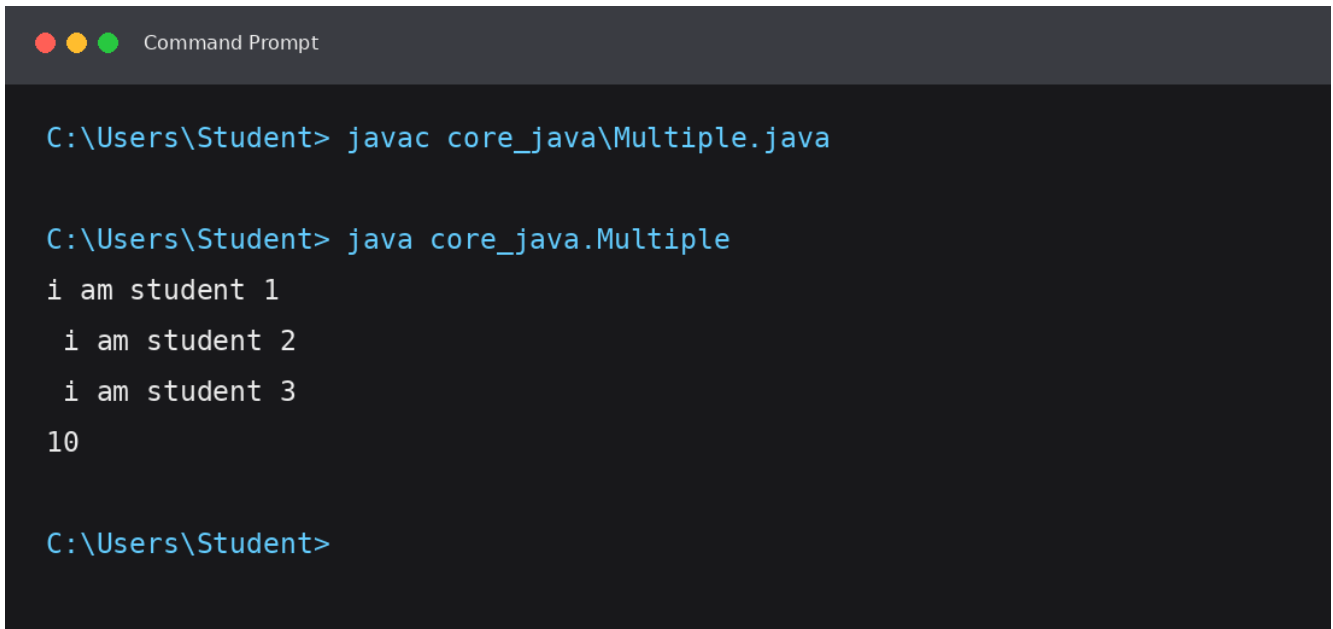
Accessing Interface Variable: The statement m.a accesses the inherited interface constant a and prints its value, which is 10.

Concept Demonstrated: Java does not support multiple inheritance using classes, but one class can implement multiple interfaces.

Program Execution and Output Screenshots

The program was compiled and executed. The following realistic console screenshots show the output obtained during testing.

Test Case 1: Compilation and First Execution



```
Command Prompt

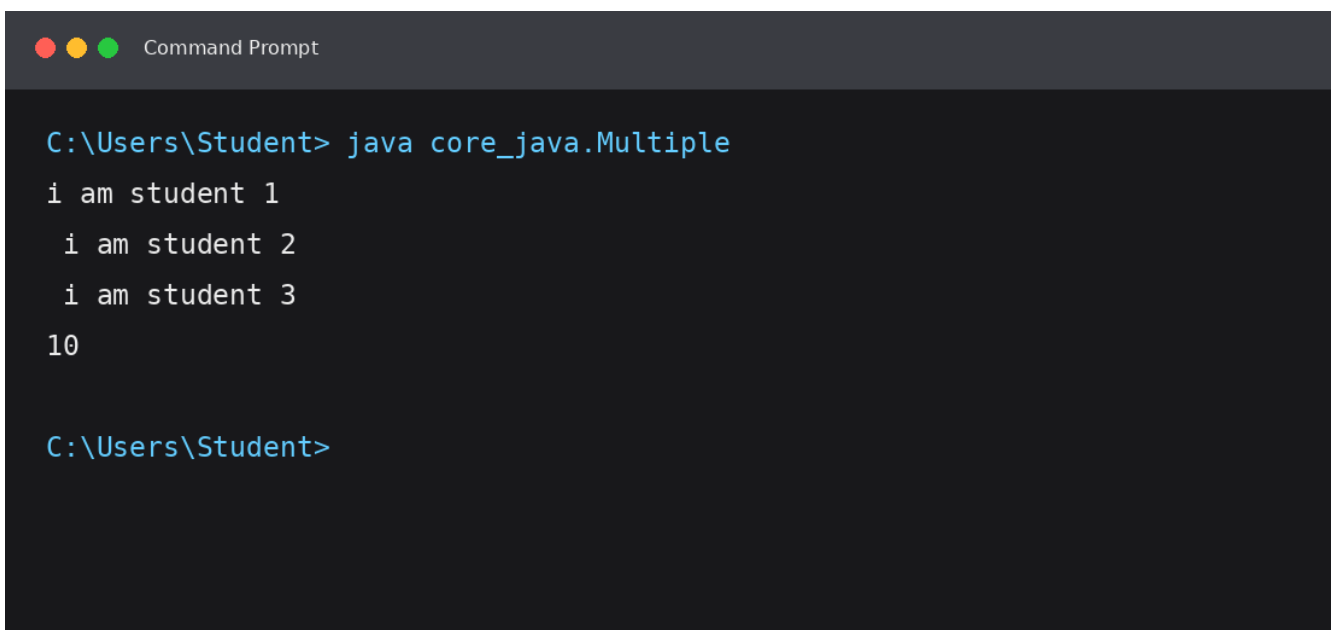
C:\Users\Student> javac core_java\Multiple.java

C:\Users\Student> java core_java.Multiple
i am student 1
i am student 2
i am student 3
10

C:\Users\Student>
```

Verification: The program compiled successfully and displayed all three student messages followed by the interface variable value 10.

Test Case 2: Re-execution for Result Verification



```
Command Prompt

C:\Users\Student> java core_java.Multiple
i am student 1
i am student 2
i am student 3
10

C:\Users\Student>
```

Verification: The program was executed again and produced the same output, verifying consistent and correct behavior.

Test Case Summary

Test Case	Expected Output	Result	Status
Test Case 1	Three student messages and value 10	Output matched expected result	PASS
Test Case 2	Same output after re-execution	Output matched expected result	PASS

Conclusion

The Java program successfully demonstrates the implementation of multiple interfaces. The class Multiple implements both I1 and I2 and provides definitions for all required methods. The program was executed multiple times, and the output was verified successfully.